

1. **DESCRIPTION:** Prior to the tournament, teams will construct, collect data on test flights, analyze and optimize free flight rubber-powered Airplanes to achieve maximum time aloft.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 15 minutes

2. **EVENT PARAMETERS:**

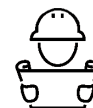
- a. Teams may bring up to 2 Airplanes, Flight Log, transportation boxes, tools, and equipment.
- b. Teams must bring one or more Measurement Boxes; transportation and measurement boxes may be the same box.
- c. The Event Supervisor will provide all other measurement tools and timing devices for scoring purposes.

3. **CONSTRUCTION PARAMETERS:**

- a. Airplanes may be constructed from published plans, commercial kits, competitor's designs, and/or other sources of design. Kits, if used, must not contain any pre-glued joints or pre-covered surfaces.
- b. Any materials except Boron filaments may be used in construction of Airplanes and boxes.
- c. The Airplane in its flight configuration, must fit fully into a team-provided Measurement Box without a lid/cover.
 - i. The external dimensions of the Measurement Box must fit within a right, rectangular prism of 32.0 cm x 24.0 cm x 47.0 cm, including any external protuberances on the lidless box.
- d. "Flight Configuration" means the Airplane is fully assembled and ready to fly. For example, no change in chord, span, length, or total lifting area can occur after removing the Airplane from its box and throughout the flight itself. Components that rotate during flight, such as propellers, may be rotated to allow the Airplane to fit into the box. The rubber motor(s) does not have to be on the Airplane or wound.
- e. Total mass of the Airplane, excluding the rubber motor(s), must be 8.0 g or more.
- f. The Propeller assemblies may be built by the participants or purchased pre-assembled. This may include a Propeller, a shaft, a hanger, and/or a thrust bearing. Variable-pitch Propellers that include mechanisms to actively change the propeller diameter or blade angle must not be used.
- g. All Airplane lifting forces must be generated by wing(s).
- h. The sole power for the Airplane must come from rubber motor(s) that remain attached to the Airplane throughout the flight.
 - i. The total mass of the rubber motor or rubber motor set, including any attachments such as O-rings, must not exceed 1.5 g.
 - ii. For a multi-motor Airplane, rubber motors must be massed together as a set, and individual rubber motors may not be interchanged between sets.
 - iii. Motors will be massed separately from the Airplane. Motors may be lubricated before and/or after check-in.
 - iv. Up to 6 motors, or sets of motors for a multi-motor Airplane, may be checked in.
- i. Participants must be able to answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org.

4. **THE COMPETITION:**

- a. Prior to day of Tournament
 - i. An indoor location will be selected by the Tournament officials. The following will be announced:
 - (1) The room dimensions (approximate length, width, and ceiling height).
 - (2) Competition factors such as air currents.
- b. Day of Tournament and prior to the first time slot:
 - i. The Event Supervisor will determine if:
 - (1) Multiple official flights may occur simultaneously.
 - (2) Practice flights may occur throughout the Competition, at the Event Supervisor's discretion, but must yield to any official flights.
 - (3) No practice flights will occur in the final half-hour of the event's last period. (Teams that declare a trim flight during their 10-minute Flight Period may still fly).
 - ii. Once participants enter the cordoned off competition area to trim, practice, or compete, they must not receive outside materials (except as permitted by the Event Supervisor), assistance, or communication. Only participants may handle the Airplane until the event ends. Teams violating this rule will be ranked below all other teams. Spectators will be in a separate area.



c. Self-Check Inspection Station (Optional)

- i. Prior to check-in with the Event Supervisor, a self-check inspection station may be made available to participants for checking their Measurement Box(es), and Airplane. Modifications may be made prior to check-in.

d. Check-In

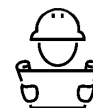
- i. At check-in, participants will present their Airplane in Measurement Box(es) and Flight Logs for inspection immediately prior to their Flight Period. Teams from the same school can share the same Measurement Box.
- ii. The Event Supervisor will verify the external dimensions of the Measurement Box(es) and that the Airplane fits fully inside the Measurement Box while in its flight configuration. Teams may not manipulate the configuration of the Airplane in order to fit into the box, except to rotate the rotor. The Airplane's overall dimensions must not change after being removed from the box. This may be verified by showing that the Airplane slides into and out of the box without changing shape at the discretion of the Event Supervisor.
- iii. The participants will remove the Airplane from the box to allow for the mass to be measured. The mass may be adjusted with ballast provided by the team to account for scale differences.
- iv. All motor(s) will be collected and returned to the team at the start of their 10-minute Flight Period.
- v. Only participants should handle the Airplane or Measurement Box(es).

e. Flight Period

- i. The 10-minute Flight Period begins when the Event Supervisor returns the motors to the team.
 - ii. Any flight beginning within the 10-minute Flight Period will be permitted to fly to completion.
 - iii. Participants may make adjustments/repairs/trim flights during their official Flight Period.
 - iv. Before each launch, participants must indicate to the Timers whether a flight is an official flight or a trim flight. A flight is considered official if a team fails to notify the Timer(s) of the flight's status.
 - v. Teams must not be given extra time to recover or repair their Airplane.
 - vi. Teams may make up to a total of 2 official flights using 1 or 2 airplanes.
 - vii. Time aloft for each flight starts when the Airplane leaves the participant's hand and stops when any part of the Airplane touches the floor, the lifting surfaces no longer support the weight of the Airplane (such as the Airplane landing on a girder or basketball hoop) or the Event Supervisors otherwise determine the flight is over.
 - viii. Event Supervisors are strongly encouraged to utilize three (3) timers on all flights. The median flight time in seconds to the precision of the device used is the official Time aloft.
 - ix. Participants must not steer the Airplane during flights.
 - x. Students must be on the floor to launch from their hands. Artificial aids cannot be used to increase launch height.
 - xi. In the unlikely event of a collision with another Airplane, a team may elect a re-flight. The decision to re-fly may be made after the Airplane lands. Timers are allowed to delay a launch to avoid a possible collision. The 10-minute Flight Period does not apply to such a flight.
- f. If requested by the Event Supervisor, the participants must demonstrate that each Airplane still fits fully inside the Measurement Box(es) in the flight configuration. Motor(s) may be removed from the Airplane or left in place during the demonstration. ES may recheck the mass of the Airplane.
- g. The Event Supervisor will review with teams their data recorded at the end of their Event Time.
- h. Teams filing an appeal must leave their Airplane(s), Motors, Measurement Box(es), and Flight Log in the event area.

5. FLIGHT LOGS:

- a. Teams may present a Flight Log of recorded data for a BONUS. This data must include 6 or more parameters (3 required and at least 3 additional) with data for 10 or more test flights prior to the competition.
 - i. The required parameters are:
 - (1) Motor Size before wind-up
 - (2) Number of Turns on Launch or Torque at Launch
 - (3) Flight Time
 - ii. The team must choose 3 additional data parameters beyond those required (e.g. turns remaining at landing, estimated/recorded peak flight altitude, the motor torque at landing, propeller pitch, etc.)



- b. All numeric data must include units which may be listed with the column headers of the log. Torque units may be “arbitrary” if an uncalibrated torque meter with arbitrary numbers is utilized.
 - c. All logs will be returned to teams after inspection.
6. **SCORING:**
- a. High Score wins.
 - b. A team’s Final Score is the larger of the team’s Flight Scores.
 - c. Flight Score for each official flight = Flight Time x Bonuses (6.d.).
 - d. **Bonuses:**
 - i. Teams with a complete Flight Log will receive a 20% bonus multiplier (x 1.2)
 - ii. Teams with a partial Flight Log will receive a 10% bonus multiplier (x 1.1).
 - iii. Teams without a Flight Log will receive no Flight Log bonus multiplier (x 1.0)
 - e. Example Score: For a flight time of 68.00 seconds with a completed Flight Log the score would be: 68.00 x 1.2 = 81.6 points.
 - f. Teams that violate rule(s) under “CONSTRUCTION PARAMETERS” or “COMPETITION” that do not have a specific penalty will be ranked after all teams that do not violate those rules.
 - g. Ties will be broken by the longest non-scored official Flight Score.
7. **DEFINITIONS:**
- a. *Airplane*: A heavier-than-air device that is held aloft by wing(s) that provide lift as the device is propelled through the air by a propeller.
 - b. *Wing*: A surface that creates lift by the flow of air across the surface.
 - c. *Event Supervisor (ES)* is responsible for ensuring the event runs correctly. The ES can assign one or more event helpers to perform various competition tasks.
 - d. *Flight Configuration* means the Airplane is fully assembled and ready to fly. For example, no change in chord, span, length, or total lifting area can occur after removing the Airplane from its box and throughout the flight itself. Components that rotate during flight, such as propellers or rotors, may be rotated to allow the Airplane to fit into the box. The rubber motor(s) does not have to be on the Airplane or wound.
 - e. *Propeller*: A rotating assembly consisting of a hub and at least one, typically two, angled and/or twisted blades designed to convert rotational energy (from rubber motor) into thrust.
 - f. *Trim Flight*: A test or practice flight that is not used for official scoring.
 - g. *Steering*: Participants using something to control the direction of motion once the device is released. (ex. Balloons, poles, rods, air, hands, etc.)
 - h. *Ballast*: A substance used to provide stability, balance, or control. (e.g. clay, tape, etc.)

This event is sponsored by the National Free Flight Society (NFFS).